



UNIVERSITY OF MALAWI - CHANCELLOR COLLEGE

Faculty of Science

Department of Mathematical Sciences

BSc. , BEd.(Science) & BSoc - Year Two

End of Semester Two Examination

MAT 221: Calculus II

29th August, 2014

Time Allowed: 3 hours

Instructions

1. This paper contains **FIVE** questions. **PLEASE CHECK.**
2. Attempt **ALL** questions.
3. Justify your working. Unsupported answers will lead to loss of marks.
4. Use of non-programmable calculators is allowed.
5. Write clearly and neatly.
6. Begin each question on a **new page**.

(Do not start answering the questions until you are told to do so)

Question One [25 Marks]

(a) Differentiate the following functions with respect to u ;

(i) $f(u) = 3u \sinh(u) \cosh(2u)$. [3 Marks]

(ii) $g(u) = u^2 \sinh^{-1}(u^2)$. [4 Marks]

(b) Show that $\int \frac{dx}{a^2 - x^2} = \frac{1}{a} \tanh^{-1}\left(\frac{x}{a}\right) + C$ for $a > 0, |x| < a$. [5 Marks]

(c) Use 1(b) to evaluate $\int \frac{x}{1 - x^4} dx$. [3 Marks]

(d) Find the length of the curve $x = \frac{1}{3}t^3, y = \frac{1}{2}t^2$ for $0 \leq t \leq 1$. [5 Marks]

(e) Find the area of the surface generated by revolving $x = t^2, y = 2t$ for $0 \leq t \leq 4$ about the x -axis. [5 Marks]

Question Two [16 Marks]

(a) Convert the cartesian equation $y = \frac{x}{x+1}$ into polar equation. [3 Marks]

(b) Transform the equation $r = 2 \cos \theta$ into rectangular coordinates. [3 Marks]

(c) Sketch and provide a name of;

(i) $r = 3 + 2 \sin \theta$. [5 Marks]

(ii) $r = \frac{10}{5 - 2 \cos \theta}$. [5 Marks]

Question Three [17 Marks]

(a) Evaluate the following limits;

(i) $\lim_{x \rightarrow 0^+} x \ln x$. [3 Marks]

(ii) $\lim_{x \rightarrow 0^+} \left(\frac{1}{e^x - 1} - \frac{1}{x} \right)$. [5 Marks]

(b) Determine whether the following improper integrals converge or diverge;

(i) $\int_{-\infty}^0 x e^x dx$. [4 Marks]

(ii) $\int_0^9 \frac{dx}{(x-1)^{2/3}}$. [5 Marks]

Question Four [24 Marks]

(a) Show that the following sequences converge to the indicated limits, L ;

(i) $\left\{ \left(\frac{1}{3} \right)^n \right\}, \quad L = 0.$ [3 Marks]

(ii) $\left\{ \left(1 + \frac{2}{n} \right)^n \right\}, \quad L = e^2.$ [6 Marks]

(b) Determine whether the series $\sum_{n=1}^{\infty} \frac{(-10)^n}{n!}$ *converges absolutely* or not. [5 Marks]

(c) Use the *root test* to determine the convergence or divergence of the series $\sum_{n=1}^{\infty} \frac{n^3}{2^n}$. [5 Marks]

(d) Use the *comparison test* to determine the convergence or divergence of the series $\frac{1}{1 \cdot 1} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots \frac{1}{n(2n-1)} + \cdots$ [5 Marks]

Question Five [18 Marks]

(a) Show that $\int_0^1 e^{-x^2} dx = 1 - \frac{1}{3} + \frac{1}{5 \cdot 2!} - \frac{1}{7 \cdot 3!} + \cdots \left[\text{Hint: } e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \right].$ [4 Marks]

(b) Use the first four terms in (a) to estimate $\int_0^1 e^{-x^2} dx$ to three-decimal-place accuracy. [2 Marks]

(c) Let $z, w \in \mathbb{C}$;

(i) Show that $|zw| = |z||w|$. [5 marks]

(ii) Show that $\overline{z+w} = \bar{z} + \bar{w}$. [4 Marks]

(d) Find the general solution to the equation $\frac{dy}{dx} = \frac{2y}{x}$ for $x > 0$. [3 Marks]

END OF QUESTION PAPER